PREDICTION METHOD OF WORDS FOR TRANSLATION OF UNKNOWN WORDS WITH UNITS EXTRACTED FROM THE PAIRS OF WORDS AND WORDS FOR TRANSLATION USING INDUCTIVE LEARNING

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Abstract

Machine translation system cannot translate words which are not registered in the dictionary. We call them unknown words. The problem that the system cannot translate unknown words is one of problems on the machine translation system. To resolve the problem, this paper proposes an automatic extraction method of units for the prediction process using inductive learning and a prediction method of word for translation with the extracted units. And we describe the results and considerations of evaluation experiment. In this experiment, the rates of effectiveness are about 60% and we confirm that our method is useful for the predictions of words for translation of the unknown words.

1 Introduction

Some approaches for machine translation have been proposed in the former researches [1, 2, 3, 4]. When we translate some text with a machine translation system, we face with unknown words of the source text. The system cannot translate these words. The reason is that the words are not registered in the dictionary of the machine translation system. Therefore we recognize the translation of unknown words as one of problems on the machine translation.

To overcome this problem, some researchers prepare the dictionary with a lot of registered words and the dictionary is used in the machine translation system. For example, the system is "Mu system [3]." They try to decrease the appearances of the unknown words with the dictionary of the machine translation system. However the problem is not resolved because the unknown words appear in the source text. One reason of the existence of unknown words is that new words are generated by the word-building capability of natural language after the dictionary was made. And the words may become unknown words. Another reason is that it is very difficult to realize the perfect dictionary which has all the words because of the heavy labor and the limitation of hardware.

In another approach to resolve the problem, the field of translation is limited. This research [4] attempts to reduce the appearances of unknown words. Therefore the system with their method can translate the text of economic news with high quality. However this problem cannot be resolved because of the

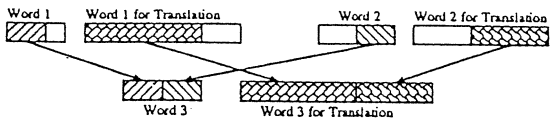


Figure 1: An illustration of prediction of word for translation with character strings extracted from the pairs of words and words for translation.

appearance of the new words. The words may become unknown words. Moreover the approach requires preparing some different dictionaries for the different fields of translation and changing the dictionaries whenever the fields of translation are changed. The labor is very heavy.

The paper [5] describes the need of process of unknown words. They do research of the appearance rates of unknown words in large corpora. And they claim that the effective clue for the inference of classification of meaning is the information of morpheme. However they does not propose the method of process for the unknown words.

To resolve the problem on the machine translation, we propose the prediction method of words for translation for unknown words. In this paper, we describe the automatic extraction method of units for the prediction process using inductive learning and the prediction method of words for translation with the extracted units. The units are extracted from two different words and two different words for translation. We call the units "a Piece of Word(:PW)" and call the pairs of PW "a Pair of Pieces of Word(:PPW)". We translate from English unknown words to Japanese words for translation.

In section 2, we explain the basic idea of our method. In section 3, we show the system based on our method. In section 4, we describe the result of evaluation experiment for our method and the consideration of the result. In section 5, our conclusions from this research are described.

2 Basic idea of Our Method

2.1 Units for Prediction Process

We think that it is possible to predict words for translation using units which are extracted from pairs of words and words for translation.

In Figure 1, a word "Word 3" and a word for translation "Word 3 for translation" are generated by some units. The units are extracted from the pair of "Word 1" and "Word 1 for translation" and the pair of "Word 2" and "Word 2 for translation".

Word 1: earthen
Word 2: oaken
⇓
String of common: en
String 1 of difference: earth
(Extracted from string 1)
String 2 of difference: oak
(Extracted from string 2)
⇓

A variable is given to the character string of common.

PW 1: @1 en
PW 2: earth
PW 3: oak

Figure 2: Examples of English PW's.

We consider that the units are extracted using inductive learning. We think that the units are PW's and that the pairs of the English and Japanese units are PPW's. The PW's and PPW's are acquired from character strings of common and character strings of difference in two words and two words for translation.

Figure 2 shows examples of English PW's. In Figure 2, "en" is the character string of common between "earthen" and "oaken." When PW's are extracted, a variable is given to the string of common at the position of strings of difference in the word. Furthermore the position of variable can substitute other PW in the prediction process. In this paper, "@1" expresses a variable. In Figure 2, PW which is extracted from the string of common is "@1 en." The strings of difference become also PW's. However a variable is not given to the strings of difference. In Figure 2, PW's which are extracted from the strings of difference is "earth" and "oak."

Figure 3 shows examples of Japanese PW's. Japanese phonics is expressed by italic characters. "@1 で作られた (@1 *de tsukura reta*)" is PW. The PW is extracted from the sting of common between "土で作られた (*Tsuchi de tsukura reta*)" and "オーク材で作られた (*Ohkuzai de tsukura reta*)." "土 (*Tsuchi*)" and "オーク材 (*Ohkuzai*)" are PPW's. The PPW's are extracted from the strings of difference between "土で作られた (*Tsuchi de tsukura reta*)" and "オーク材で作られた (*Ohkuzai de tsukura reta*)."

Figure 4 shows examples of PPW's. "@1 en, @1 で作られた (@1 *de tsukura reta*)" is PPW. The PPW is the pair of English and Japanese PW. The English PW is extracted from the sting of common between two English words. And the Japanese PW is extracted from the character string of common between two Japanese words for translation. "earth, 土 (*Tsuchi*)" and "oak, オーク材 (*Ohkuzai*)" are PPW's.

String 1: 土で作られた (*Tsuchi de tsukura reta*)
String 2: オーク材で作られた (*Ohkuzai de tsukura reta*)
⇓
String of common: で作られた (*de tsukura reta*)
String 1 of difference: 土 (*Tsuchi*)
(Extracted from String 1)
String 2 of difference: オーク材 (*Ohkuzai*)
(Extracted from String 2)

⇓

A variable is given to the character string of common.

PW 1: @1 で作られた (*de tsukura reta*)
PW 2: 土 (*tsuchi*)
PW 3: オーク材 (*Ohkuzai*)

Figure 3: Examples of Japanese PW's.

The PPW's are also the pairs of English and Japanese PW. The English PW's are extracted from the stings of difference between two English words. And the Japanese PW's are extracted from the string of difference between two Japanese words for translation.

Furthermore, Figure 5 shows an example of prediction with PPW's. The prediction result "silken, 絹で作られた (*Kinu de tsukura reta*)" is compounded from PPW "@1 en, @1 で作られた (*@1 de tsukura reta*) and PPW "silk, 絹 (*Kinu*)". And the result is equal to the correct word for translation.

2.2 Field of Our Research

Our research aims to resolve the problem that the machine translation system cannot translate the unknown words. The machine translation system requires the words for translation which adapts to the context of source text. When the words are selected, the system requires some informations. For example they are a thesaurus and so on. However, when the selection process of the word for translation is executed, the required informations may be unregistered in the source of information. We think that there are some problems in the process for the selection of the words for translation, which adapt to the context.

Therefore we recognize the process that the system selects the most adaptable words as other problems to resolve in the near future. In this research, the system based on our method predicts some words for translation which may be suitable.

Pair 1:

Word 1: earthen
Word 1 for translation: 土で作られた (*Tsuchi de tsukura reta*)

Pair 2:

Word 2: oaken
Word 2 for translation: オーク材で作られた (*Ohkuzai de tsukura reta*)
⇓
Pair of stirings of common: en で作られた (*de tsukura reta*)
Pair 1 of strings of difference: earth 土 (*Tsuchi*)
(Extracted from Pair 1)
Pair 2 of strings of difference: oak オーク材 (*Ohkuzai*)
(Extracted from Pair 2)
⇓

A variable is given to the character string of common.

PPW 1: @1 en @1 で作られた (*de tsukura reta*)
PPW 2: earth 土 (*Tsuchi*)
PPW 3: oak オーク材 (*Ohkuzai*)

Figure 4: Examples of PPW's.

3 Outline of Our method

3.1 Overview

Figure 6 shows the outline of system based on our method.

After the input of an unknown word, the prediction process is executed. At the beginning of the prediction process, the system searches for the PPW's which have the English PW's which constructs the appearance of written the unknown word. Thereafter, the Japanese PW's of the searched PPW's generated

Unknown English word: silken
PPW 1: @1 en @1 で作られた (*de tsukura reta*)
PPW 2: silk 絹 (*Kinu*)
⇓
Result : silken 絹 で作られた (*Kinu de tsukura reta*)
Correct word for translation: silken 絹 で作られた *Kinu de tsukura reta*

Figure 5: An Example of prediction with PPW's

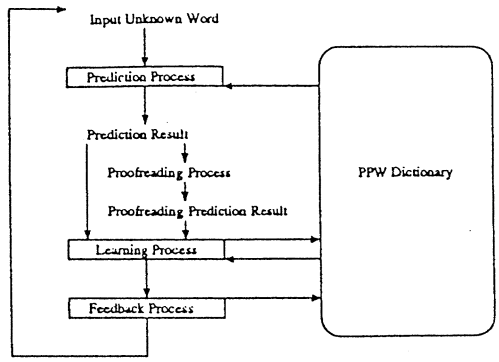


Figure 6: Outline of system based on our methods.

the results of prediction. When two Japanese PW's are combined, in the case that one PW or both PW have a variable, the PW is substituted for the variable in another PW.

New PPW's are acquired from the pairs of English words and Japanese words for translation and PPW's which were acquired using inductive learning before this process. The PW's are extracted from the string of common and the strings of difference between two source character strings. The new pairs of extracted English and Japanese strings become new PPW's. The new PPW's are added to PPW dictionary.

After the judgment that the prediction result is whether correct or erroneous, the numerical values of PPW's are calculated in the feedback process. The calculations in the case that the result is correct are different from the calculation in the case that the result is erroneous. The differences in this feedback process may make the system to select better PPW's on the prediction process. Therefore the process improves the capability of prediction.

3.2 Predction Process

On this process, the system searches PPW's whose English PW's compose the appearance of written unknown word and it generates the result of prediction with the Japanese PW's of searched PPW's.

However in the case that there are some available combinations of Japanese PW's, some prediction results are generated by them. In such a case, the system should decide the order of priority among the results and inform the user of the certain result. When the system ranks the results in order of certainly, it refers to some numeric values of PPW's and Japanese PW's. The numeric values are as follows:

1. the number of used PPW's in the prediction result.
2. the average of appearance frequency of PPW's used in the prediction result.
3. the average of correct prediction frequency of PPW's used in the prediction result.
4. the average of erroneous prediction frequency of PPW's used in the prediction result.
5. the average of correct prediction frequency of Japanese PW used in the prediction result.
6. the average of erroneous prediction frequency of Japanese PW used in the prediction result.

The appearance frequency of PPW expresses how many times the PPW has been used in the prediction process. The correct prediction frequency of PPW indicates how many times the PPW has been used in the correct prediction results and the erroneous prediction frequency of PPW indicates how many times the PPW has been used in the erroneous prediction results. Furthermore the correct connection frequency of Japanese PW shows how many times the connection has been used in the correct prediction results and the erroneous connection frequency of Japanese PW shows how many time the connection has been used in the erroneous prediction results.

3.3 Learning Process

The given PPW's are insufficient to predict words for translation of unknown words. In the learning process new PPW's are extracted from pairs of English words and Japanese words for translation and from PPW's which are registered in the PPW dictionary.

When the new PPW's are acquired, the system uses with the information of the appearance of written words and words for translation. Therefore the system are extracted some PPW's which have erroneous correspondences of meaning between English PW's and Japanese PW's. To reduce the erroneous correspondences, we define some condition for the learning process. The conditions are as follows:

Condition 1 when English PW's are extracted, the character strings of common are more than two characters.

Condition2 when PPW's are extracted,

Condition 2-a one source pair of English and Japanese strings is composed by one pair of common character strings and another is composed by one pair of common character strings and one source pairs of different character strings.

Pair 1:

Word 1:	gentleman
Word 1 for translation:	紳士 (*Shinsi*)

Pair 2:

Word 2:		gentlemanly
Word 2 for translation:		紳士的な (*Shinsi tekina*)
	⇓	
PPW 1:	gentleman @1	紳士 @1 (*Shinsi*)
PPW 2:	ly	的な (*tekina*)

Figure 7: Examples of PPW's adapted to Condition 2-(a)

Pair 1:

Word 1:	thermometer
Word 1 for translation:	温度計 (*Ondo kei*)

Pair 2:

Word 2:		voltmeter
Word 2 for translation:		電圧計 (*Denatsu kei*)
	⇓	
PPW 1:	@1 meter	@1 計 (*kei*)
PPW 2:	thermo	温度 (*Ondo*)
PPW 2:	volt	電圧 (*Denatsu*)

Figure 8: Examples of PPW's adapted to Condition 2-(b)

Condition 2-b both of pairs are composed by one string of common and one string of difference.

Figure 7 shows that an example of PPW extraction is applied to Condition 2-a. And Figure 8 shows taht an example of PPW extraction is applied to Condition 2-b.

3.4 Feedback Process

On the feedback process, the process in the case that the system predicted the correct result is different from the process in the case that the system predicted the erroneous result. The process makes the system to select more certain PPW on the prediction process and improves the capability of prediction process. The certainty values are treated. The value of 1 is added one. In the case that the prediction result is correct, the values of 2-(a) and (b) are added one. And in the case that the prediction result is erroneous, the values of 3-(a) and (b) are added one.

The values are as follows:

1. the appearance frequency of all PPW's used in the prediction result
2. in the case that the prediction result is correct
   (a) the correct prediction frequency of PPW's used in the prediction result
   (b) the correct connection frequency of Japanese PW used in the prediction result
3. in the case that the prediction result is erroneous
   (a) the erroneous prediction frequency of PPW's used in the prediction result
   (b) the erroneous connection frequency of Japanese PW used in the prediction result

The definitions of the values are described in the above section.

4 Evaluation Experiment

4.1 Initial PPW Dictionary

The pairs of the bases and their words for translation are defined as PPW's. The reason is that the pairs are extracted from words and words for translation. In this paper, we extracted the pairs from the pairs of words and words for translation which are registered in the dictionary [6]. The dictionary has 49,336 English words and may be useful for people to read general text. We extracted 102,156 pairs of English words and their Japanese words for translation from the dictionary.

Furthermore, the pairs of affixes and their words for translation are also defined as PPW's in this paper. The reason is that the pairs are extracted from words and their words for translation. We consulted the book [7] to extract the pairs of affixes and their words for translation. As a result of the extraction, 351 PPW's are extracted. They are included 69 PPW's with English PW's which have the same character string of prefix and Japanese words for translation. And they are also included 282 PPW's with English PW's which have the same character string of suffix and Japanese words for translation.

Figure 9 shows examples of PPW's which are extracted from the pairs of English words with the same

Pair 1:

Word 1:	ultramodern
Word 1 for translation:	超現代的な (*Chou gendai tekina*)

Pair 2:

	Word 2:	ultrashort
	Word 2 for translation:	超短波の (*Chou tanpa no*)
	⇓	
PPW 1:	ultra @1	超 @1 (*Chou*)
PPW 2:	modern	現代的な (*gendaite kina*)
PPW 3:	short	短波の (*tanpa no*)

Figure 9: Examples of PPW's extracted from the pairs of words with the same prefix and words for translation

Pair 1:

Word 1:	endless
Word 1 for translation:	終りのない (*Owari no nai*)

Pair 2:

	Word 2:	worthless
	Word 2 for translation:	価値のない (*Kachi no nai*)
	⇓	
PPW 1:	@1 less	@1 のない (*no nai*)
PPW 2:	end	終り (*owari*)
PPW 3:	worth	価値 (*kati*)

Figure 10: Examples of PPW's extracted from the pairs of words with the same postfix and words for translation

prefix and Japanese words for translation. In Figure 9, PPW 1 is added to the initial PPW dictionary. And Figure 10 shows examples of PPW's which are extratced from the pairs of English words with the same postfix and Japanese words for translation. In Figure 10, PPW 1 is added to the initial PPW's.

In this experiment, the total number of initial PPW's is 102,507. And they are added to the PPW dictionary of the system before the experiment.

4.2 Definition of Unknown Words

In this experiment, unknown words are recognized as the words which are not registered in the dictionary [6]. The kinds of unknown words are derivatives and compound words. Therefore we selected the data for the the experiment extracted from the unknown words in the corpus. The experiment data are extracted from four categories in Susanne corpus [8]. These categories are "Press Reportage," "Bells letters, Biography, Memories," "Learned Writing" and "Adventure and Western Fiction." Table 1 shows the numbers of words, the numbers of different words, the numbers of different unknown words and the numbers of experiment data in the four categories.

In this table, (A) expresses the numbers of words, (B) expresses the numbers of different words, (C) expresses the numbers of different unknown words, and (D) expresses the numbers of experiment data in the categories.

4.3 Procedure of Evaluation Experiment

Before the experiment, 102,507 PPW's which were described above were added to PPW's dictionary before the experiment. The experiment data are experimented. They are executed with the prediction process, the proofreading process in the case that the result is erroneous, the learning process and the feedback process. The outline of the system is described in Figure 6 .

4.4 Standard of Evaluation

The results of prediction are classified into three groups. We define that the correct words for translation of the unknown words are registered in the English-Japanese dictionary [6]. They are described below.

Group 1 The result is equal to the appearance of written the correct word for translation or the only one part in the result is rearranged by the equivalent sting.

Group 2 The result is equal to a part of the appearance of written the correct word for translation.

Group 3 The result cannot classified neither Group 1 nor 2.

Figure 11 shows an example of prediction result of Group 2. The prediction result "正当なに (*Seitou na ni*)" is equal to a part of the appearance of written correct words for translation"正当な (*Seitou na*)".

Table 1: Experiment data

Name of category	(A)	(B)	(C)	(D)
Press Reportage	39,243	4,425	1,119	179
Belles Letters, Biography, Memories	39,062	4,986	845	305
Learned Writing	37,880	3,851	739	370
Adventure and Western Fiction	40,399	3,896	580	202

Unknown English word: rightfully
PPW 1: @1 ly @1 に (*ni*)
PPW 2: rightful 正当な (*Seitou na*)
⇓
Result : rightfully 正当な に (*Seitou na ni*)
Correct word for translation: rightfully 正当に (*Seitou ni*)

Figure 11: An Example of result of Group 2

Furthermore we define the rate of effectiveness. The value expresses the effectiveness in this experiment. The total number of experiment data divides the total number of Group 1 and Group 2. The expression of the rate is as follows:

the rate of effectiveness(%) =

$$\frac{(Group\quad 1)+(Group\quad 2)}{(experiment\quad data)} * 100$$

4.5 Results

Table 2 shows the results of this evaluation experiment. In Table 2, (A) expresses the numbers of experiment data (B) expresses the numbers of Group 1, (C) expresses the numbers of Group 2, (D) expresses the rates of effectiveness and (E) expresses the numbers of Group 3 in the categories.

In Table 2, the rates of effectiveness in the four categories are 63.1%, 63.0%, 62.1% and 66.3%. The system cannot know the words for translation of the experiment data because it is not registered in the dictionary. However, with our method, the rates of effectiveness are about 60% and the system can predict some correct words for translation among the experiment data. Therefore we confirm that our method is useful to predict words for translation of unknown words.

Table 3: Numbers and rates of Group 1-1 in categories

Name of category	(A)	(B)	(C) (%)
Press Reportage	26	20	76.9
Belles Letters, Biography, Memories	41	27	65.9
Learned Writing	50	21	42.0
Adventure and Western Fiction	20	14	70.0

4.6 Considerations

4.6.1 Discussion on Group 1

In this section, we discuss the results of Group 1. Based on the source of ecause of the classification of the PPW's which compose the result, the results of Group 1 are divided into three groups. The three groups are described below.

Group 1-1 The results are composed by PPW's which are in the initial PPW dictionary.

Group 1-2 The results are composed by both of PPW's which are in the initial PPW dictionary and PPW's which are extracted in the learning process.

Group 1-3 The results are composed by PPW's which are extracted in the learning process.

Table 3 shows the numbers and rates of the Groups 1-1. In Table 3, (A) expresses the numbers of Group 1, (B) expresses the numbers of Group 1-1 and (C) expresses the rates of Group 1-1 to Group 1 in the categories. Figure 12 shows an example of the result of Group 1-1. The PPW's are given to the PPW dictionary of the system before the experiment.

Table 4 shows the numbers and rates of the Groups 1-2. In Table 4, (A) expresses the numbers of Group 1, (B) expresses the numbers of Group 1-2 and (C) expresses the rates of Group 1-2 to

Table 2: Results of evaluation experiment

Name of category	(A)	(B)	(C)	(D)	(E)
Press Reportage	179	26	87	63.1	66
Belles Letters, Biography, Memories	292	41	143	63.0	108
Learned Writing	370	50	180	62.1	139
Adventure and Western Fiction	202	20	114	66.3	68

Unknown English word: redevelopment

PPW 1:	re @1	再 @1 (*Sai*)
PPW 2:	re @1	再び @1 (*Futatabi*)
PPW 3:	development	開発 (*kaihatsu*)
PPW 2:	development	発達 (*hattatu*)
PPW 2:	development	発展 (*hatten*)
		etc.
	⇓	
Results :	redevelopment	再開発 (*Sai kaihatsu*)
		再発達 (*Sai hattatsu*)
		再発展 (*Sai hatten*)
		再び開発 (*Futatabi kaihatsu*)
		再び発達 (*Futatabi hattatsu*)
		再び発展 (*Futatabi hatten*)
		etc.
Correct word for translation:		redevelopment 再開発 (*Sai kaihatsu*)

Figure 12: Examples of prediction result of Group 1-1

Group 1 in the categories. Figure 13 shows an example of the result of group 1-2. The PPW:"thermo @1, 熱 @1 (*Netsu*)" is extracted from both of the pair: "thermocouple, 熱電対 (*Netsu denn tsui*)" and the pair:"thermodynamic-equilibrium, 熱平衡 (*Nestu heikou*)." And the PPW:"dynamic, 力学の (*Rikigaku no*)" is regicstered in the initial PPW dictionary.

Table 5 shows the numbers and rates of the Groups 1-3. In Table 5, (A) expresses the numbers of Group 1 (B) expresses the numbers of Group 1-3 and (C) expresses the rates of Group 1-3 to Group 1 in the categories. Figure 14 shows an example of Group 1-3. The PPW:"right @1, 右 @1 (*Migi*)" is extracted from both of the pair:"right-handler, 右ききの人 (*Migi kiki no hito*)" and the pair:"right, 右 (*Migi*)." And the PPW:"-field, 翼 (Yoku)" is extracted from both of the pair:"left-field, 左翼 (*Sayoku*)" and the pair:"left, 左 (*Hidari*)."

Table 4: Numbers and rates of Group 1-2 in categories

Name of category	(A)	(B)	(C)(%)
Press Reportage	26	5	19.2
Belles Letters, Biography, Memories	41	9	22.0
Learned Writing	50	16	32.0
Adventure and Western Fiction	20	4	20.0

Table 5: Numbers and rates of Group 1-3 in categories

Name of category	(A)	(B)	(C) (%)
Press Reportage	26	1	3.8
Belles Letters, Biography, Memories	41	5	12.2
Learned Writing	50	13	26.0
Adventure and Western Fiction	20	2	10.0

In Figure 4 and 5, the rates in the category "Learned Writing" are higher than other categories. In this category, the numeric values indicate that the system acquired more PPW's for the prediction process in this category than others. And the PPW's are effective. Some technical terms appear in the text of this category and the appearances of words have some resemblance. The system based on our method uses the information of the appearance of written words. Therefore it can acquire the effective PPW's in the case of this. There are 21 technical terms in 50 words of Group 1 and 132 technical terms in 370 words of

Unknown English word: thermodynamic

PPW 1:	thermo @1	熱 @1 (*Netsu*)
PPW 2:	thermo @1	熱電 @1 (*Netsuden*)
PPW 3:	dynamic	力学の (*rikigaku no*)
PPW 2:	dynamic	動力の (*douryoku no*)
PPW 2:	dynamic	活動的な (*katsudou tekina*)
		etc.

⇓

Results :	thermodynamic	熱力学の (*Netsu rikigaku no*)
		熱動力の (*Netsu douryoku no*)
		熱活動的な (*Netsu katudou tekina*)
		熱電力学の (*Netsuden rikigaku no*)
		熱電動力の (*Netsuden douryoku no*)
		熱電活動的な (*Netsudenn katsudou tekina*)
		etc.

Correct word for translation: thermodynamic
熱力学の
(*Sai kaihatsu*)

Figure 13: Examples of prediction result of Group 1-2

Unkown English word: right-field

PPW 1:	right @1	右 @1 (*Migi*)
PPW 2:	right	正しい (*Tadasii*)
PPW 3:	@1 - @2	@1 @2
PPW 4:	-field	翼 (*Tsubasa*)
PPW 5:	field	野原 (*Nohara*)
		etc.

⇓

Results :	thermodynamic	右翼 (*Uyoku*)
		正しい野原 (*Tadasii nohara*)
		etc.

Correct word for translation: right-field
右翼
(*Uyoku*)

Figure 14: Examples of prediction result of Group 1-3

the experiment data of this category.

The rates of Group 1-2 and 1-3 in the category "Press Reportage" are lower than other categories. The category is abundant in topics. The examples of the topics are a baseball, a politics, a court case and so on. Therefore the number of effective PPW's for prediction process is a few number. However the rate of effectiveness in this category is almost equal to the rate of effectiveness in other categories. The reason is that the system used PPW's which are in the initial PPW dictionary in the prediction process. Therefore the system can indicate almost the fixed rate of effectiveness.

4.6.2 Discussion on Group 2

Because the result is classified into Group 2, we classified Group 2 into three groups. The groups are as follows:

Group 2-1 the shortage of a necessary variable in the English PW for the prediction process

Group 2-2 the shortage of one necessary PPW's for the prediction process in the PPW dictionary

Table 6: Numbers and rates of Group 2-1 in categories

Name of category	(A)	(B)	(C)(%)
Press Reportage	87	13	14.9
Belles Letters, Biography, Memories	143	13	9.1
Learned Writing	180	12	6.7
Adventure and Western Fiction	114	8	7.0

Group 2-3 unequal to a part of the appearance of written correct word for translation

Table 6 shows the numbers and rates of Group 2-1 in the Group 2. In Table 6, (A) expresses the numbers of Group 2, (B) expresses the numbers of Group 2-1 and (C) expresses the rates of Group 2-1 to Group1 in the categories.

Table 7 shows the numbers and rates of Group 2-2 in the Group 2. In Table 7, (A) expresses the numbers of Group 2, (B) expresses the numbers of Group 2-2 and (C) expresses the rates of Group 2-2 to Group1 in the categories.

Table 8 shows the numbers and rate of Group 2-3 in the Group 2. In Table 8, (A) expresses the numbers of Group 2, (B) expresses the numbers of Group 2-3 and (C) expresses the rates of Group 2-3 to Group1 in the categories.

Table 7: Numbers and rates of Group 2-2 in categories

Name of category	(A)	(B)	(C) (%)
Press Reportage	87	44	50.6
Belles Letters, Biography, Memories	143	57	39.9
Learned Writing	180	76	42.3
Adventure and Western Fiction	114	38	33.3

Table 8: Numbers and rates of Group 2-3 in categories

Name of category	(A)	(B)	(C) (%)
Press Reportage	87	30	34.5
Belles Letters, Biography, Memories	143	73	51.0
Learned Writing	180	92	51.1
Adventure and Western Fiction	114	68	59.7

An example of Group 2-1 is the prediction result of unknown word "committeewoman." The word for translation of unknown word"committeewoman" is "女性委員 (*Josei iin*)." In the prediction process, the system found some PPW's in the PPW dictionary. There are PPW:"committee, 委員 (*iin*)" and PPW:"woman, 女性 (*Josei*)" in the PPW dictionary. However these PPW's do not have a variable in the PPW's and the prediction result can not be generated. In the case that there is a PPW"committee @1, @1 委員 (*iin*)," the system can generate the result "女性委員 (*Josei iin*)." And the result is equal to the correct word for translation.

The prediction result of unknown word: "semicircular"is classified into Group 2-2. The word for translation of unknown word"semicircular" is "半円形の (*Han enkei no*)." In the prediction result, the system found PPW:"circular, 円形の (*Enkei no*)" and there is not PPW:"semi @1, 半 @1(*Han*)" in the PPW dictionary. Therefore the system can not generate the result which is equal to the correct word for translation.

Figure 11 shows an example of the result of Group 2-3. It is not equal to the ending of the correct word for translation.

Table 9: Numbers and rates of Group 3-1 in categories

Name of category	(A)	(B)	(C)(%)
Press Reportage	66	39	59.1
Belles Letters, Biography, Memories	108	62	57.4
Learned Writing	138	91	65.9
Adventure and Western Fiction	68	35	51.4

In the category of "Press Reportage", the rates of Group 2-1 and 2-2 are higher than others. The reason is that the text in this category is abudant in topics and the similarities of the appearance of the written word is a few similarities. The system uses the information of the appearance of written words. Therefore the number that the system extracted PPW's is a few number.

To change the results of Group 2-1 and 2-2 for the better, we should improve the system with our method of the learning process. And the system should be able to acquire more effective PPW's.

4.6.3 Discussion on Group 3

Because the system generates the erroneous results, we grouped Group 3 into three groups. The groups are as follows:

Group 3-1 the shortage of PPW's for the prediction process

Group 3-2 the correct word for translation that can not be generated by the PPW's

Group 3-3 the difference of the appearance of written words

Table 9 shows the numbers and rates of the Groups 3-1. In Table 9, (A) expresses the numbers of Group 3, (B) expresses the numbers of Group 3-1 and (C) expresses the rates of Group 3-1 to Group 3 in the categories.

Table 10 shows the numbers and rates of the Groups 3-2. In Table 10, (A) expresses the numbers of Group 3, (B) expresses the numbers of Group 3-2 and (C) expresses the rates of Group 3-2 to Group 3 in the categories.

Table 11 shows the numbers and rates of the Groups 3-3. In Table 11, (A) expresses the numbers of Group 3, (B) expresses the numbers of Group 3-3 and (C) expresses the rates of Group 3-3 to Group 3 in categories.

Table 10: Numbers and rates of Group 3-2 in categories

Name of category	(A)	(B)	(C)(%)
Press Reportage	66	26	39.4
Belles Letters, Biography, Memories	108	36	33.3
Learned Writing	138	40	29.0
Adventure and Western Fiction	68	32	47.1

Table 11: Numbers and rates of Group 3-3 in categories

Name of category	(A)	(B)	(C) (%)
Press Reportage	66	1	1.5
Belles Letters, Biography, Memories	108	10	9.3
Learned Writing	138	8	5.8
Adventure and Western Fiction	68	1	1.5

The rates of Group 3-1 in four categories are high. To acquire more effective PPW's in our method, the system requires the source character strings which have the high similarity to other strings on the appearance of written strings.

An example of Group 3-2 is the prediction result of the unknown word:"strong-room." And the word for translation of it is "金庫室 (*Kinko shitsu*)." The correct word for translation cannot generated by PPW's. The Japanese PW of English PW:"room" can be PW: "室 (*shitsu*)." However the Japanese PW of English PW:"strong" can not be PW:"金庫 (*Kinko*)." The word for translation cannot be generated by PPW's. Therefore the system cannot predict the words for translation. We consider that we should give the words for translation to the system.

An example of Group 3-3 is the prediction result of unknown words:"colourful." In general, we spell "colorful" of the word. The system uses the information of the appearance of wrritten words. In the case that the appearance of written one character string is different from the appearance of written another character string, the prediction process with PPW's is ineffective. To recover this problem, we should discuss the use of another informations. The examples of the informations are the rules of spelling and so on.

5 Conclusion

This paper describes the extraction method of PPW's for the prediction process using inductive learning and the prediction method of the unknown words with PPW's. And we describe the results and considerations of the evaluation experiment with the system based on our method. The machine translation system cannot know the words for translation of unknown words. However, from the results of the evaluation experiment, the rates of effectiveness are about 60%. Therefore, we confirm that our method is useful for the prediction of the unknown words.

The machine translation system requires the adaptable words for translation to the context of text. We will limit the field of sources of PPW's to the words for translation, which adapt to the context of the text. And we will try to predict the adaptable words for translation in the future work.

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